Unified Data Library (UDL)
Cross Mission Data Branch

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Highest Overall Classification:
UNCLASSIFIED

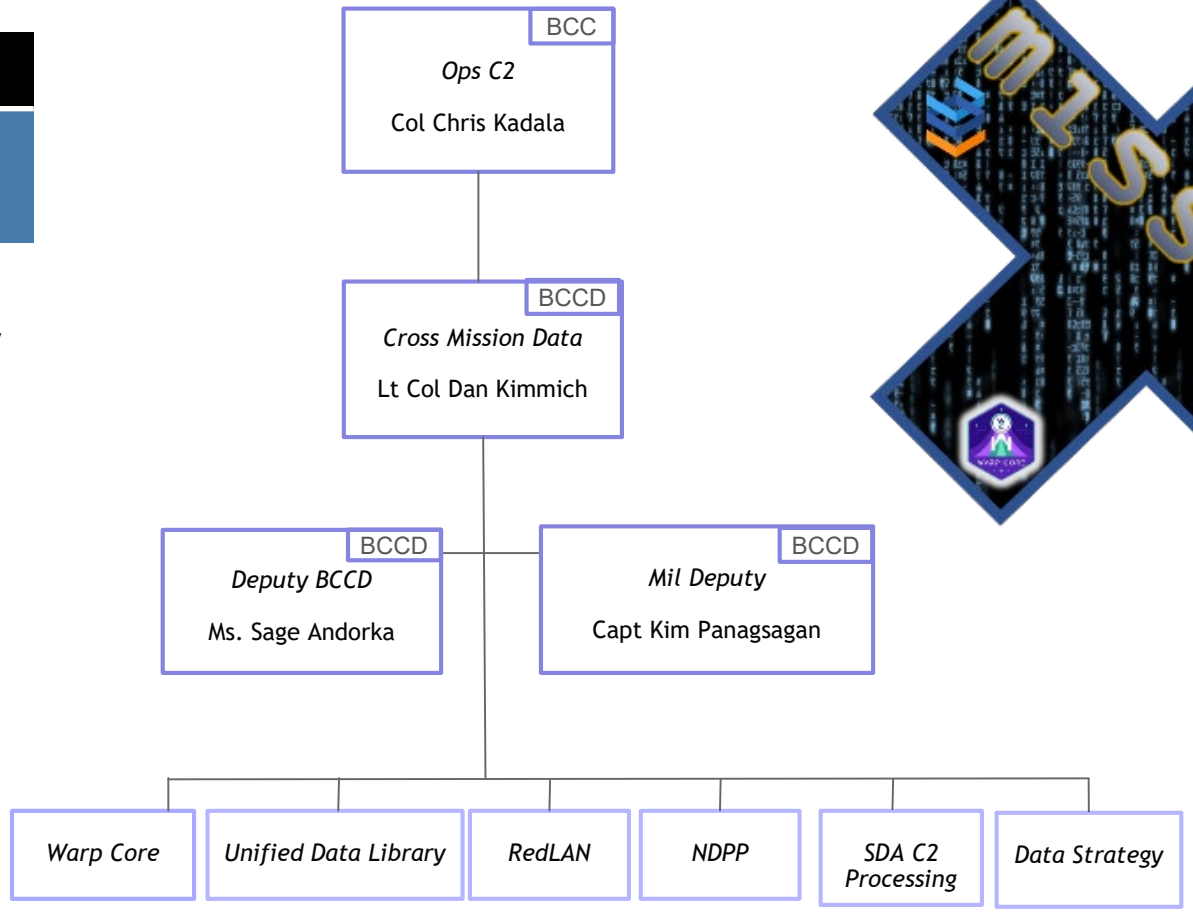
Cross Mission Data Division

CROSS MISSION GROUND & COMMUNICATIONS ENTERPRISE

Cross-Mission Data Mission:

Provide data solutions that enable Command and Control at the speed of battle.

- The creation of the Cross Mission Data branch is an explicit response to the Chief of Space Operations (CSO) intent to break down barriers between stove-piped systems across the space enterprise
- Vision: Enable decision dominance at the speed of relevance
- Core Competencies: Ingestion, Storage, Exposure, Discovery, Distribution; Translation & Pre-processing; Provenance; Data visualization; Cross-Domain Solutions; Federation



Programmatically Aligns Key SSC Data Initiatives

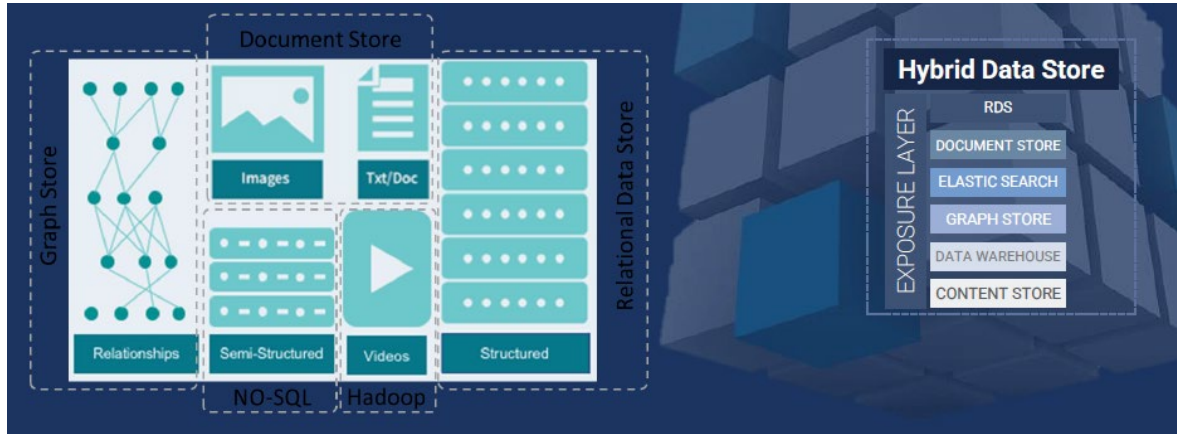
Bottom Line Up Front

- The UDL continues to deliver the foundation for data management across the USSF, leading the Services in data exposure and exploitation
- The UDL institutes the principles of zero-trust and is proven to be highly reliable, durable, and cyber secure across classification levels
- Architecture capabilities planned for 2022 will enable data delivery to allies
- The UDL is an extensible and adaptable Government owned system, capable of evolving to meet future requirements
- The UDL is a cornerstone of the DAF data fabric

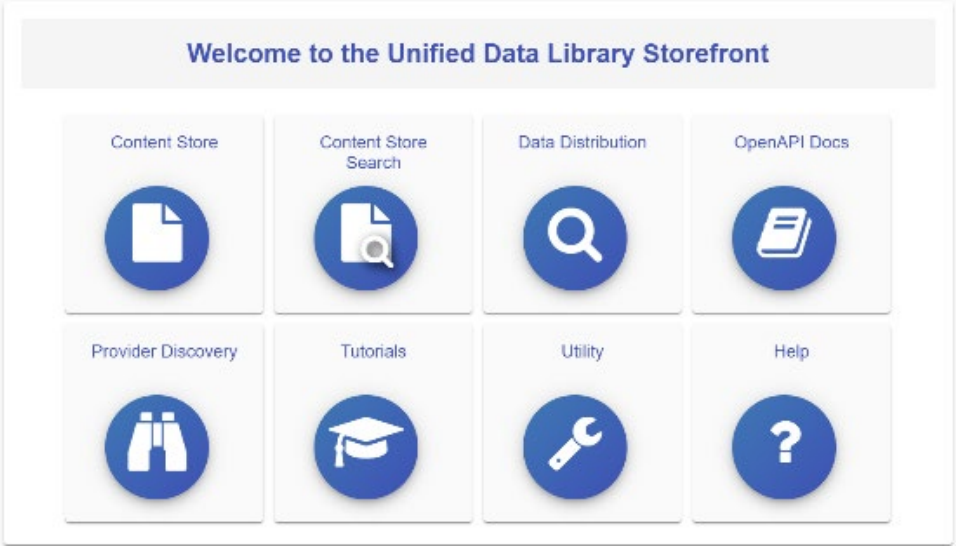
The UDL is the SDA Data Repository to Win the 2026 Fight

Current State

- Current user base consists of 150 academic, 1700 commercial, 125 allied and 2000+ government accounts
  - Common data store front-end built on best-of-breed and industry leading commercially licensed software
  - Currently satisfies 90K requests every hour as normal operations
  - 30 subscription topics & 300+ supporting schemas for different data types
  - Successfully integrated the first sensor directly to the UDL - Space Fence



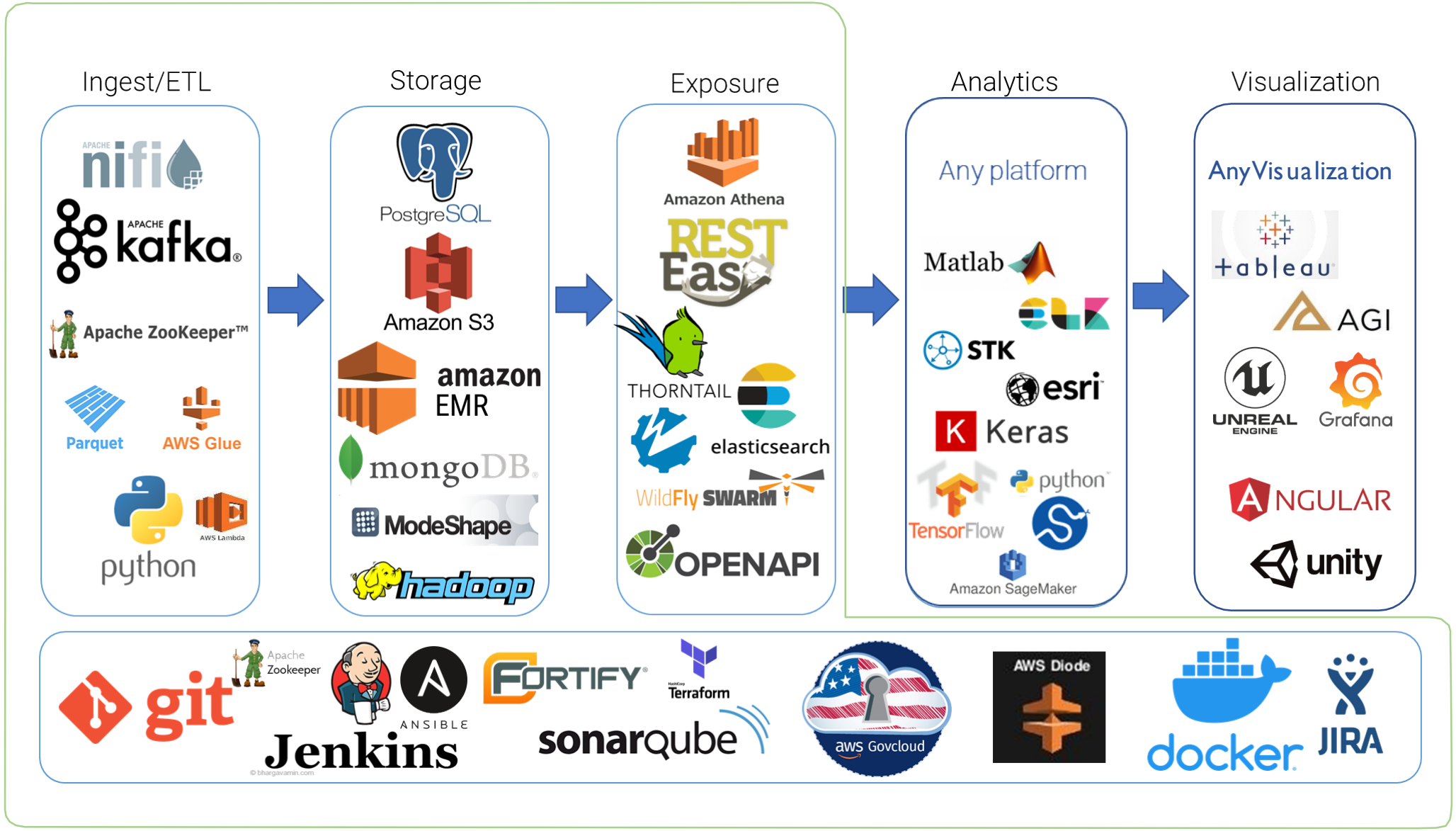
UDL Storefront



Unclass: https://unifieddatalibrary.com (Unclassified)
SIPR: https://unifieddatalibrary.af.smil.mil (SIPR)
JWICS: https://unifieddatalibrary.af.ic.gov (JWICS)

- **Data Distribution can be used to query the UDL**
- **Open API Docs provides information on the data parameters accessible in the UDL and how to code to post or query for information**
- **Provider Discovery has a "Baseball Card" for each data provider. You can select each one and there are additional cards for each of their data types. If you are looking to access any provider's data, you will have to "request access" for each data type you are interested in. Note that many of the commercial providers require a contract (money) to provide you their data, but there are several government sources that are "free"**
- **Tutorials contain some short videos on various aspects of the UDL**
- **Help provides more detailed examples and support for the various UDL functions**

UDL Open Technology Stack



UDL Capabilities

The UDL broadly exposes data from closed architectures

Directly supports the SDA mission:

- Space data schemas were developed with the SDA community
- Manages commercial SDA data for the USSF enterprise
- Directly connected to Space Fence, the most sensitive search radar in the Space Surveillance Network (SSN), the Department of Defense (DoD) network of ground and space-based sensors that provide observation data on objects in the space environment

Structured to support multiple mission areas across the USSF

- Sets Space Force data modeling standards & baseline schema/fields for data ingestion & exposure
- Agnostic to visualization & analytic capabilities
- Controls & delivers data across multiple security enclaves
- Singular framework for data discovery, normalization & distribution
- Authority to Operate (ATO) at Unclassified, Secret and Top Secret levels

An Aerial view of the Space Fence facility located on Kwajalein Atoll

Why We Need the UDL

- The UDL delivers value by broadly exposing data from closed architectures
  - The UDL sets Space Force data modeling standards & baseline schema/fields for data ingestion & exposure
  - Space data schemas were developed alongside SDA community, reducing data consumer & provider costs
  - Effectively manages commercial SDA data for the USSF enterprise
  - The UDL platform is agnostic to visualization & analytic capabilities
- Controls & delivers data across multiple security enclaves
  - Singular framework for data discovery, normalization & distribution

Enterprise Data Exposure Foundation for USSF to Evolve Decision-Making

Backup Slides

Purpose of the UDL

- Data agnostic cloud-based enterprise data repository, free the data!
- Discovery, Ingest, Secure, Storage, and Access to data at scale
- Provide logically centralized and geographically distributed
- Permit machine-to-machine multi-domain operations
- Ensure proper data management and secure data distribution
- Eliminate the N^2 integration problem of point to point
- Reduce duplication of effort in data integration across programs

Customers

UNIFIED DATA LIBRARY

- The UDL User-base
  - 150 academic, 1700 commercial, 125 allied, and 2000+ government accounts across 25 countries;
  - 300+ schemas, APIs provide international data standards for space traffic management
  - Supports 12 programs of record
  - 70 applications pulling in UDL data with 15 million observations pulled/day
  - Supported the integration of track data in support of OAR real-world Afghanistan operations
  - UDL is the foundational data layer for JTF-SD Commercial Operations worldwide with coalition partners and commercial providers

LINES OF EFFORT

Platform	•Backbone of UDL with internal CI/CD pipeline and 24/7 team
Space Onboarding	•Establishes data standards for sharing and prioritizes community data needs for onboarding through steering committee
SOCOM	•Integration of SOF Mission Data Layer with tactical sources (TAK/TRAX federation, GCCS-J, MARS, MIDB, GDELT, Link-16) and single sign on capability
Astro-Standards	•Securely distribute and track access to protected Astrodynamic Standards using UDL security processes
SSDP	•Mod and Sim support and future SAP development
AFRL	•DevSecOps pipeline for Space Force and JTF-SD
SDA Marketplace	•19 providers across 5 countries, supports international transactions, RFP to TS/SCI data flow demonstrated in 17 minutes. 90+ commercial data acquisitions awarded in first 12 months
RCO/ABMS	•Operational level UDL, in-theater access to minimize latency with data replication to enterprise UDL. DataONE for ABMS, and adopted by USAFE, PACAF, and NORAD/Northcom; Integrated 70 data sources in 3 months with a team of 6 developers
CENTCOM/SHIELD	•Tactical Edge connectivity, allows all theater radars to operate as independent IOT devices, S//REL UDL development (AxE) leveraging E-UDL to allow 2-way real time operational data sharing between US and ALL coalition data partners at the SECRET levels, endorsed by multiple COCOMs
Tactical UDL	•Purpose built to connect to tactical air, ground, and maritime platforms. Provides centralized log management and software depot for software updates and cyber patches to all globally deployed UDL instances
JTF/SD	•Supports protect and defend mission; provide dedicated developer support, classified data prioritization, rapid onboarding, and foundation for JCO and SACT events
Air Mobile Command	•Afghan Evacuation, Air Data Integration, GDSS connectivity, Puckboard Crew Papers, aircraft sensor data, aircraft messaging, all integrated within 72 hours
Army Data Fabric Layer	•Integrates graph store for knowledge graph curation, builds automated data ingestion utilities, and demonstrates federation across services
Intelligence Community	•Silent Barker leveraging UDL for enterprise distribution of payload data, integration w/ 3 letter agency enables first ever DoD, IC, and commercial colocation of data

UDL Recent Successes

- Supported ABMS on-ramps and Sprint Advanced Concept Training (SACT) demonstrations; integrated key data sets to assist in mission demonstrations, to include support for the Russian Direct Ascent (DA) Anti-satellite (ASAT) and Shijian-21 activities
- During the Project Convergence 2021 (PC21) experiment, the UDL supported multiple scenarios in which air-and-missile defense engagements were conducted in response to simulated enemy missile attacks. This was accomplished by completing the Sensor-To-Shooter kill chain by rapidly connecting sensors to ingest and share data through the UDL to Army weapon systems for a live fire execution
- AMC/GDSS – Integrated and distributed OAR/OAW mission data from GDSS to authorized users across the USG without the need for new devices or software at the user end. Blue Heron COP, FBI, DHS, N/NC and other government agencies were utilizing the data to track refugee movements OCONUS to CONUS minutes to hours after the data went live
- Early discovery and engineering efforts with the Department of the Air Force Rapid Capabilities Office (DAFRCO) to develop a UDL outpost across data center geographical regions in a peer-to-peer paradigm that does not rely on a master node responsible for keeping all peer nodes in synchronization.

UDL Enables Whole-of-Government Response – Space leading the Global Fight

Data Security

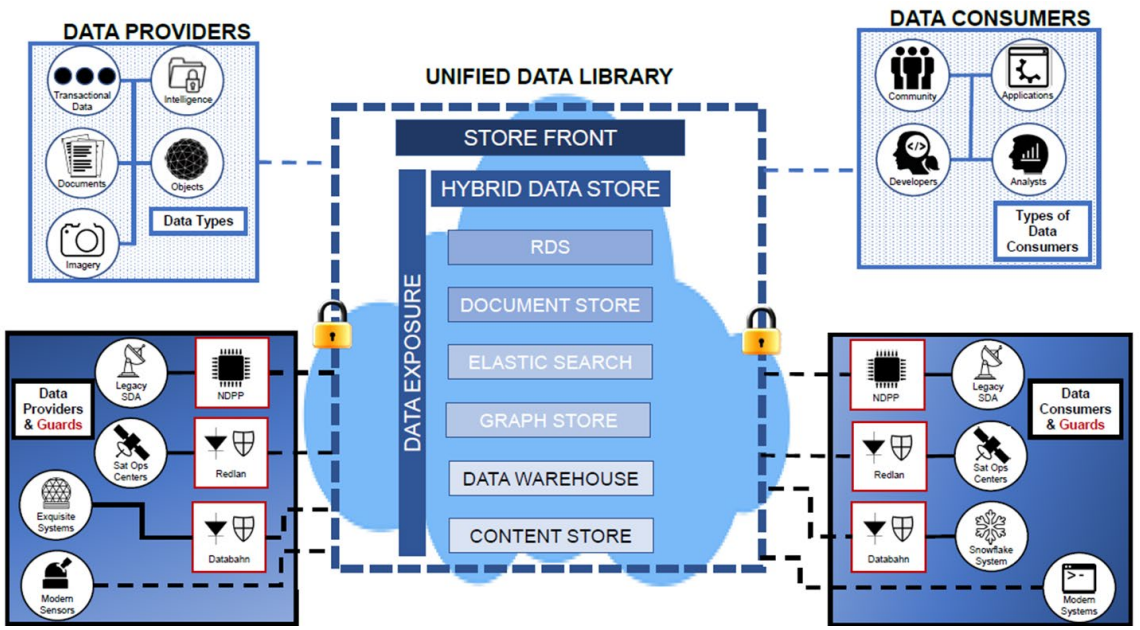
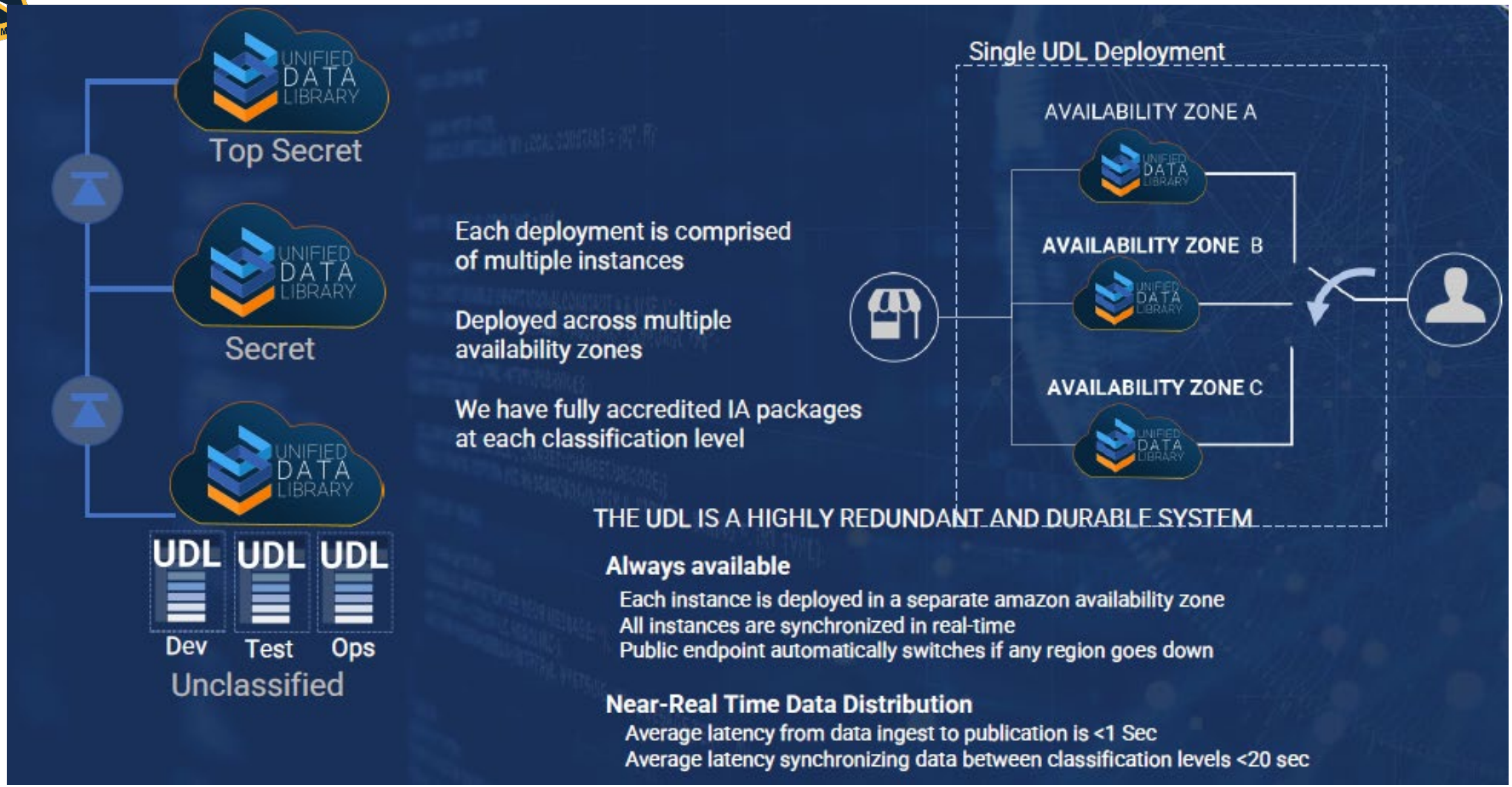
- 3-yr ATO issued for all enclaves in April 2022, first among hundreds of programs
- Completed Cooperative Vulnerability Penetration Assessment (CVPA) completed with 47th Cyberspace Test Squadron
  - Dark Wolf - Red team penetration testing; weekly tag up & monthly campaigns
  - UDL provides Dashboard results, auditing all user activity with event notifications & monitoring in real-time

> "The test team is struggling to find any vulnerabilities"
> – 47th Cyberspace Test Squadron, 8 Dec 21

- Fine-grain access controls enabled via data and attribute policies
  - Security tags must be submitted appropriately by the data provider to be accepted
  - The UDL assesses each data service call, determining if that specific record can be delivered to the user as part of the service response
  - Access can be controlled by object, time-window, classification, service or data element

UDL Performance

- MIT Lincoln Laboratory contracted by the SPO in August 2021 to perform a baseline characterization of UDL performance
- Latency was measured in the sub-second range, with an average latency of 144 milliseconds
- UDL meets user performance requirements
  - The team has focused on delivering a secure, easy to use, fault tolerant, and highly availability system (Ao 99.9%)
  - The UDL is deployed across multiple AWS availability zones for redundancy & failover
  - The development backlog includes high performance database ingestion and pub/sub capability



Performance & Resilience